NautiCAI
AI-Powered Underwater Infrastructure Inspection

CORROSION VELOCITY MODEL — TIME-TO-FAILURE REPORT

1. SYSTEM OVERVIEW

NautiCAI Corrosion Velocity Model tracks defect growth rate across multiple inspection cycles and uses polynomial regression to predict exactly when a defect will become critical. Output: a per-asset time-to-failure risk score. This moves NautiCAI from inspection tool to risk management platform.

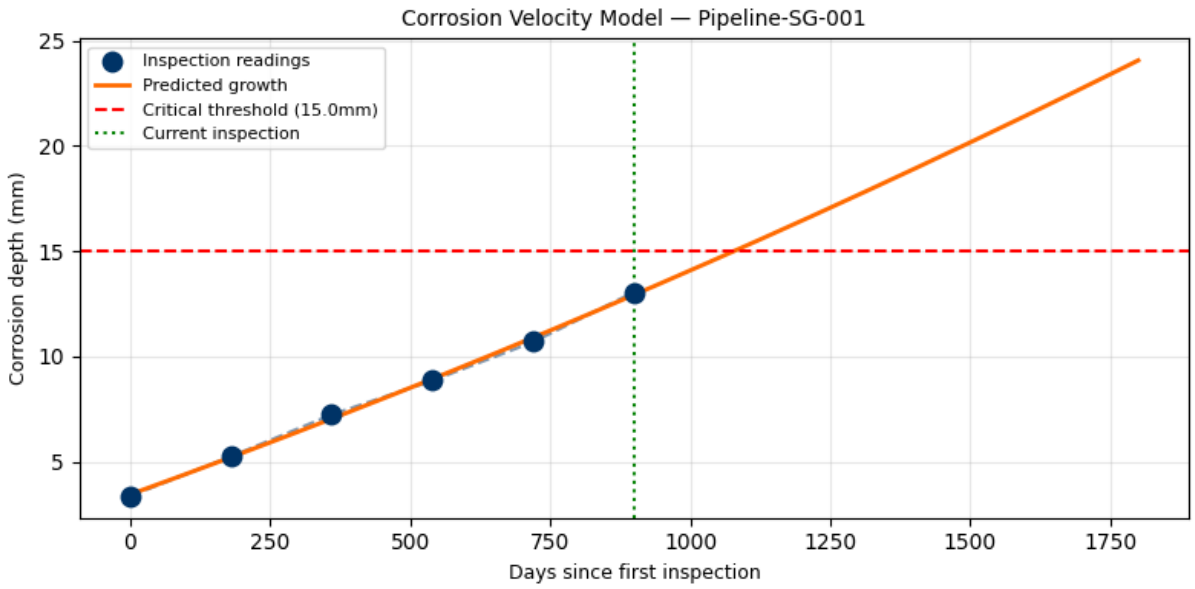
2. ASSET RISK SUMMARY

Asset ID	Current Corrosion	Growth Rate	Time to Failure	Risk Level
Pipeline-SG-001	13.05 mm	0.3 mm/month	6 months	HIGH
Pipeline-SG-002	10.99 mm	0.2 mm/month	18 months	MODERATE
Pipeline-SG-003	10.59 mm	0.2 mm/month	24 months	LOW
Hull-FR-001	10.3 mm	0.2 mm/month	33 months	LOW
Hull-FR-002	10.37 mm	0.2 mm/month	33 months	LOW

3. DETAILED ASSET ANALYSIS

Asset: Pipeline-SG-001 — Risk: HIGH

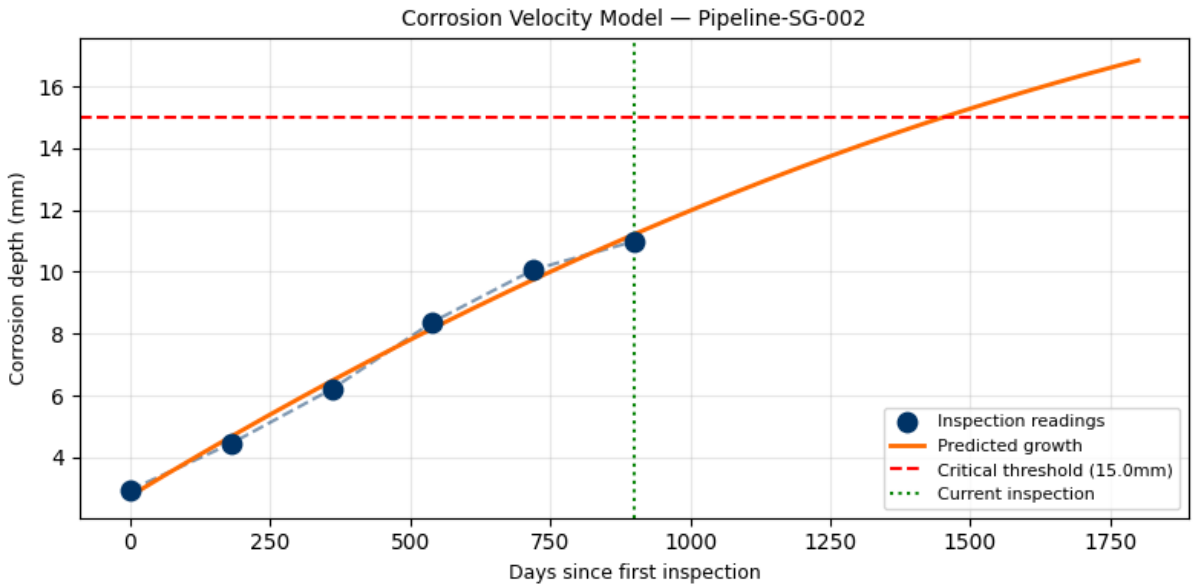
Cycle	Date	Corrosion (mm)	AI Confidence
1	Jan 2022	3.37	90%
2	Jun 2022	5.24	86%
3	Dec 2022	7.23	88%
4	Jun 2023	8.85	83%
5	Dec 2023	10.69	86%
6	Jun 2024	13.05	83%



Predicted time to critical threshold: 6 months | Growth rate: 0.3 mm/month

Asset: Pipeline-SG-002 — Risk: MODERATE

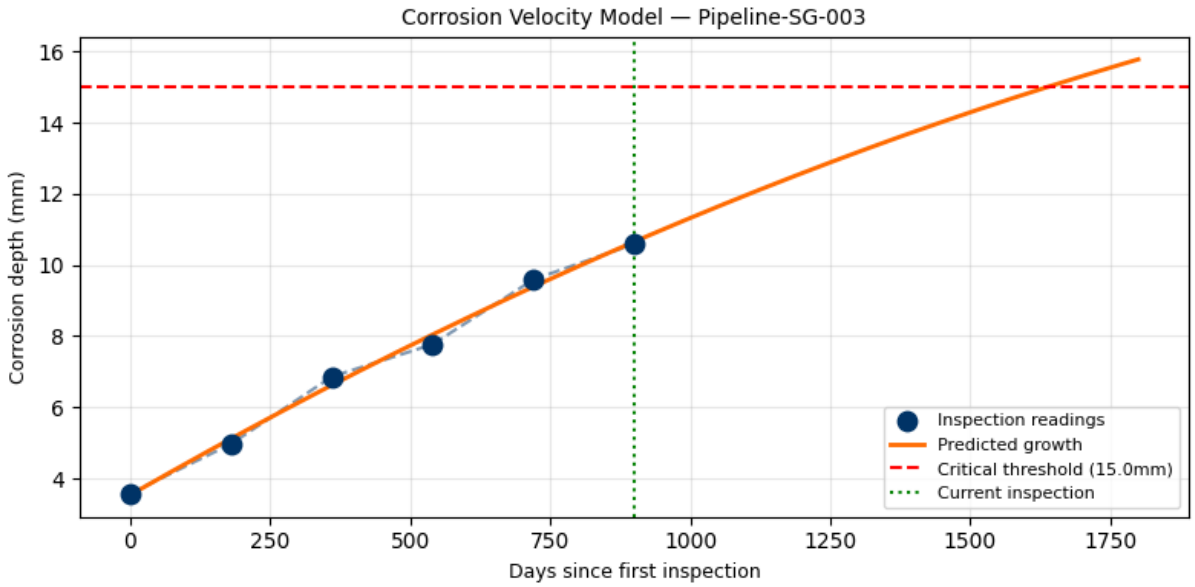
Cycle	Date	Corrosion (mm)	AI Confidence
1	Jan 2022	2.95	85%
2	Jun 2022	4.45	87%
3	Dec 2022	6.21	90%
4	Jun 2023	8.38	96%
5	Dec 2023	10.07	88%
6	Jun 2024	10.99	95%



Predicted time to critical threshold: 18 months | Growth rate: 0.2 mm/month

Asset: Pipeline-SG-003 — Risk: LOW

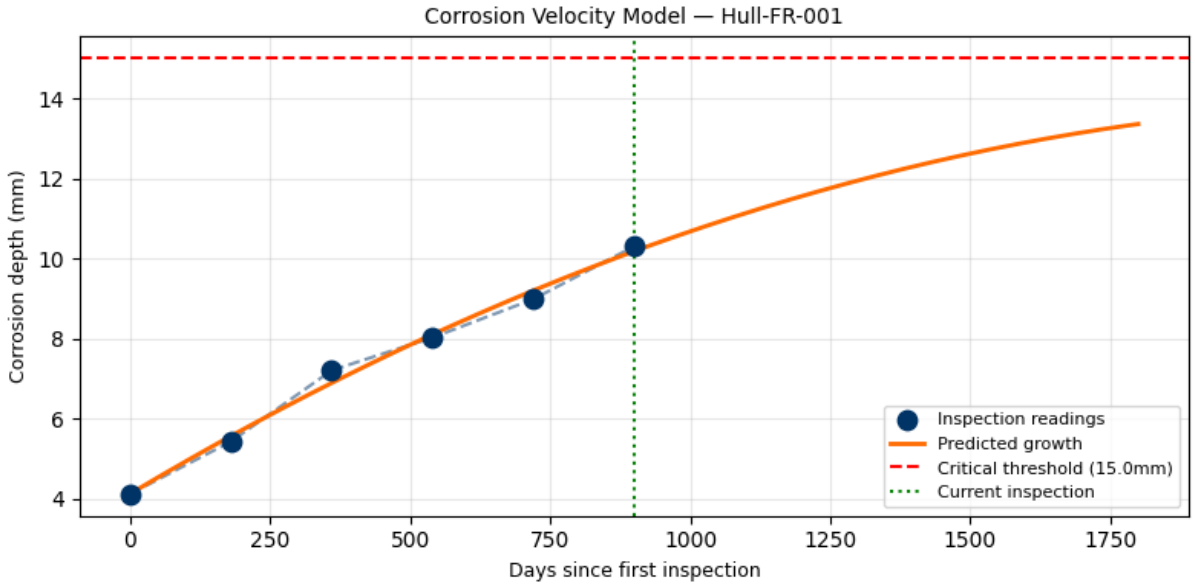
Cycle	Date	Corrosion (mm)	AI Confidence
1	Jan 2022	3.58	91%
2	Jun 2022	4.97	87%
3	Dec 2022	6.86	86%
4	Jun 2023	7.77	92%
5	Dec 2023	9.59	84%
6	Jun 2024	10.59	96%



Predicted time to critical threshold: 24 months | Growth rate: 0.2 mm/month

Asset: Hull-FR-001 — Risk: LOW

Cycle	Date	Corrosion (mm)	AI Confidence
1	Jan 2022	4.11	89%
2	Jun 2022	5.42	84%
3	Dec 2022	7.2	88%
4	Jun 2023	8.02	93%
5	Dec 2023	8.98	96%
6	Jun 2024	10.3	82%

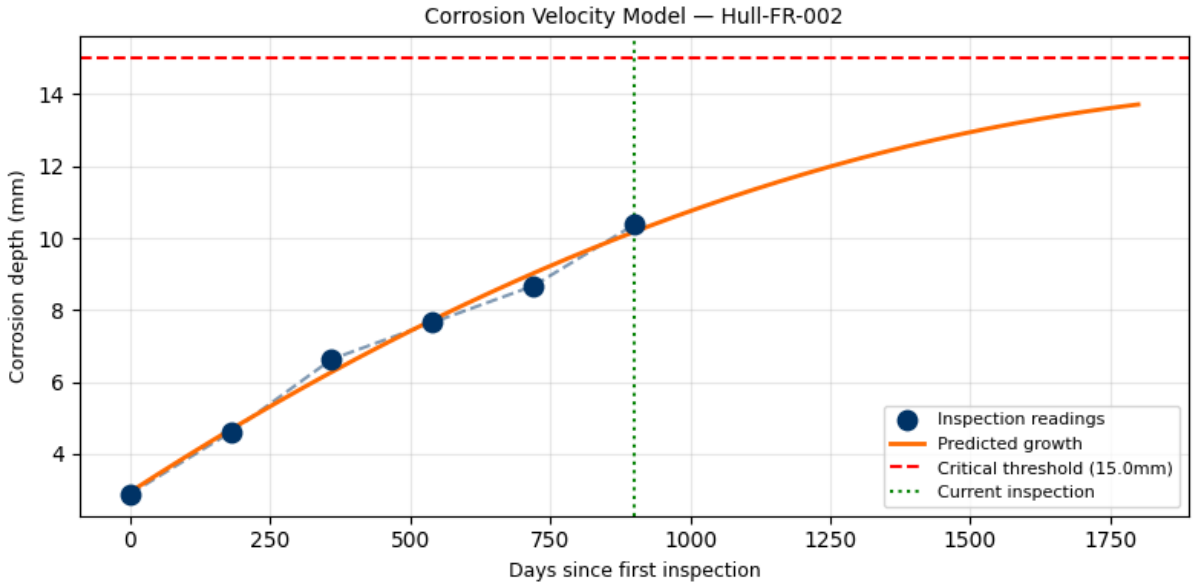


Predicted time to critical threshold: 33 months | Growth rate: 0.2 mm/month

Asset: Hull-FR-002 — Risk: LOW

Cycle	Date	Corrosion (mm)	AI Confidence

1	Jan 2022	2.88	93%
2	Jun 2022	4.6	96%
3	Dec 2022	6.63	85%
4	Jun 2023	7.65	92%
5	Dec 2023	8.67	83%
6	Jun 2024	10.37	96%



Predicted time to critical threshold: 33 months | Growth rate: 0.2 mm/month

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Corrosion velocity calculated using polynomial regression on multi-cycle YOLOv8 detection outputs. Critical threshold set at 15mm per DNV-RP-C203 guidelines. NautiCAI Singapore.